

Adam Ingah

303-883-5210 | adam.ingah@gmail.com | linkedin.com/in/adamingah | adamingah.com

Summary

U.S. Citizen based in Seattle, WA, pursuing an MSEE part-time at the University of Washington. Early-career electrical engineer with hands-on experience in embedded systems, PCB design, and IoT development, seeking opportunities in embedded and hardware engineering.

Education

University of Washington, Seattle	M.S. Electrical Engineering	June 2027
University of Alabama, Birmingham	B.S. Electrical Engineering	May 2025

- GPA: 3.65/4.00 | Dean's List (Four Semesters)
- Relevant Coursework: Machine Learning, Internet of Things

Experience

Hardware Lead | UAB Engineering Lab | 8/2023 - 5/2024

- Designed and prototyped a low-cost assistive vision device for visually impaired users, reducing cost by ~90% compared to commercial solutions
- Led hardware testing and iterative prototyping in partnership with engineers and patients, refining usability and accuracy through feedback-driven design cycles.

Lab Assistant | UAB Libraries | 8/2022 - 5/2025

- Maintained and repaired 5+ 3D printers, enabling 20+ student projects throughout the semester with minimal printer downtime.
- Collaborated with student engineers and researchers during design reviews, offering technical troubleshooting and CAD guidance to improve project outcomes.

Projects

Autonomous Fire Extinguishing Robot | adamingah.com/projects/pages/amfr.html

- Built an autonomous robot using Raspberry Pi 4 and thermal vision to detect and extinguish fires in real time
- Led hardware integration and rapid prototyping with 3D-printed components, ensuring reliable sensor-actuator communication under test conditions.

IoT Plant Irrigation System | adamingah.com/projects/pages/ipws.html

- Developed an ESP32-based irrigation system with real-time soil monitoring and automated pump control, reducing manual watering needs.
- Integrated analog sensors with Arduino IoT Cloud via UART/SPI/I2C protocols, enabling remote data tracking and control.

Low Vision Magnifier | adamingah.com/projects/pages/lvm.html

- Designed a text magnifier with 3D-printed housings and Raspberry Pi-based OCR (OpenCV + Tesseract) for individuals with low vision.
- Delivered an affordable, modular, and accessible solution compared to commercial devices.

Skills

Programming: C, Assembly, Java, Python

Hardware: ESP32, Arduino, Raspberry Pi,

Tools: LTSpice, Altium Designer, Git, SolidWorks, Fusion 360, MATLAB,